Exercise 9

On curve studying

From the school book | Understand | Apply | Higher Order Thinking Skills

First Problems on curve sketching

1 **Sketch the curves of the functions defined by the following rules :**

(1) $f(x) = 3 - x^2$

(2) $f(x) = x^2 - 6x + 5$

(3) $f(x) = 12x - x^3$

(4) $f(x) = x(x+3)^2$

(5) $f(x) = 2x^3 - 9x^2 + 12x - 2$

(6) $f(x) = x^3 - 6x^2 + 12x - 5$

(7) $f(x) = (2-x)(x+1)^2$

(8) $f(x) = x(x-2)^2 + 3$

(9) $f(x) = \frac{1}{8}(x+4)(x-2)^2$

(10) $f(x) = x|x-4|$

(11) $f(x) = \begin{cases} x^3 - 3x^2 & , x > 0 \\ x^2 - 2x & , x \le 0 \end{cases}$

2 *(1st Session 2012)* Determine the intervals of increasing and decreasing for the curve $y = x^3 - 3x + 4$, then sketch the graph of the curve indicating on it the points of local maximum , local minimum and the points of inflection (if exist). « $(-1, 6)$, $(1, 2)$, $(0, 4)$ »

3 *(2nd Session 2004)* Find the local maximum and the local minimum values of the function f , where $f(x) = (x+1)(x^2 - x - 2)$ as well as the inflection points of its curve (if they exist) and finally sketch its graph. « zero , -4 , $(0, -2)$ »

4 *(1st Session 2004)* For the function f , where $f(x) = x^3 - 3x^2 + 4$, find the local maximum and minimum values as well as the points of inflection of its curve (if they exist) in the interval $]-1, 3[$, then sketch the graph of this function over $[-1, 3]$ « 4 , 0 , $(1, 2)$ »

5 **Prove that the curve whose equation :**

$y = x^3 + x^2 + x$ has no local maximum or local minimum , then sketch the curve.

6 **Sketch the curve of the function f giving that :**

(1) f is continuous and its domain $[1, 3]$ | (2) $\grave{f}(x) > 0$ for every $x \in [1, 3]$

(3) $\grave{\grave{f}}(x) < 0$ for every $x \in [1, 3]$

7 📖 **Sketch the curve of the continuous function f where :**

(1) $f(0) = 4 \; f(3) = 4$

(2) $\dot{f}(x) < 0$ for each $x < 2$, $\dot{f}(x) > 0$ for each $x > 2$

(3) $\ddot{f}(x) > 0$

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8 **Sketch the curve of the continuous function if :**

(1) $f(3) = 5$

(2) $\dot{f}(3) = 0$

(3) $\ddot{f}(x) < 0$ for each x

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9 **Sketch the curve of the function f where $y = f(x)$ if :**

(1) f is continuous and its domain is $[-1 , 5]$

(2) $f(1) = f(3) = 0$, $\dot{f}(2)$ is not existed.

(3) $\dot{f}(x) > 0$ when $x < 2$, $\dot{f}(x) < 0$ when $x > 2$

(4) $\ddot{f}(x) > 0$ when $x \neq 2$

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10 **Sketch the curve of the continuous function f in which :**

(1) $f(3) = 5$, $\dot{f}(3) =$ zero

(2) $\ddot{f}(x) > 0$ for each $x < 3$, $\ddot{f}(x) < 0$ for each $x > 3$

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11 📖 **Sketch the curve of the function f where $y = f(x)$ if you know the following :**

(1) f is continuous whose domain is $[0 , \infty[$, $f(4) = 3$, $f(0) = 1$

(2) $\dot{f}(x) > 0$ when $x > 0$

(3) $\ddot{f}(x) > 0$ when $x < 4$, $\ddot{f}(4) = 0$, $\ddot{f}(x) < 0$ when $x > 4$

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12 📖 **Sketch the curve of the continuous function f which has the following properties :**

(1) $f(0) = 3$ | (2) $\dot{f}(2) = \dot{f}(-2) = 0$

(3) $\dot{f}(x) > 0$ when $-2 < x < 2$

(4) $\ddot{f}(x) < 0$ when $x > 0$, $\ddot{f}(x) > 0$ when $x < 0$

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13 **Graph the curve of the continuous function f which satisfies the following properties :**

(1) $f(4) = 2 \; f(3) = 4$, $f(2) = 0$

(2) $\dot{f}(x) < 0$ when $x > 4$ or $x < 2$

(3) $\ddot{f}(x) < 0$ when $x > 3$, $\ddot{f}(x) > 0$ when $x < 3$

14 📖 **Graph the curve of the continuous function f which satisfies the following properties :**

(1) $f(-1) = 2\ f(0) = 4$, $f(1) = 0$

(2) $\grave{f}(1) = \grave{f}(-1) = 0$

(3) $\acute{f}(x) < 0$ for each $x < 0$, $\acute{f}(x) > 0$ for each $x > 0$

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15 📖 Determine a , b such that the point $(1 , 3)$ is an inflection to the curve $y = a x^3 + b x^2$, then sketch the curve.

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16 Identify the values of a , b , c and d to the curve $y = a x^3 + b x^2 + c x + d$ where it has a local maximum value at $(0 , 6)$ and a local minimum value at $(1 , 5)$, then sketch the curve.

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17 Let the curve $y = x^3 + a x^2 + b x$ have an inflection point at $(-1 , 11)$

(1) Find the equation of the curve « $y = x^3 + 3 x^2 - 9 x$ »

(2) Determine the points of local maximum and local minimum. « $(-3 , 27)$, $(1 , -5)$ »

(3) Sketch the curve.

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18 If $f(x) = x^3 + a x^2 + b x + 4$ where a and b are two constants. Find the two values of a and b if the function f has a local minimum value when $x = 2$ and an inflection point when $x = 1$, then sketch the curve of the function f « $-3 , 0$ »

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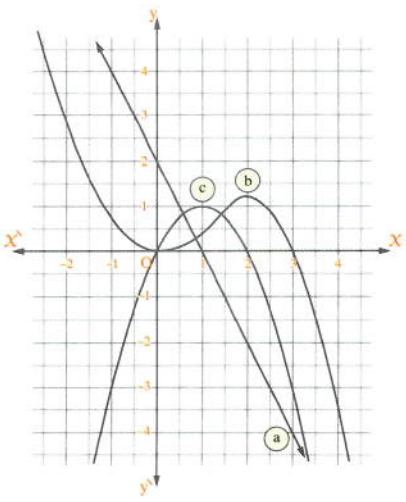
19 **Find the absolute maximum value and absolute minimum value of the function f where :**

$$f(x) = \begin{cases} x + 1 & , \ x < 1 \\ (x - 3)^2 - 2 & , \ x \geq 1 \end{cases}$$

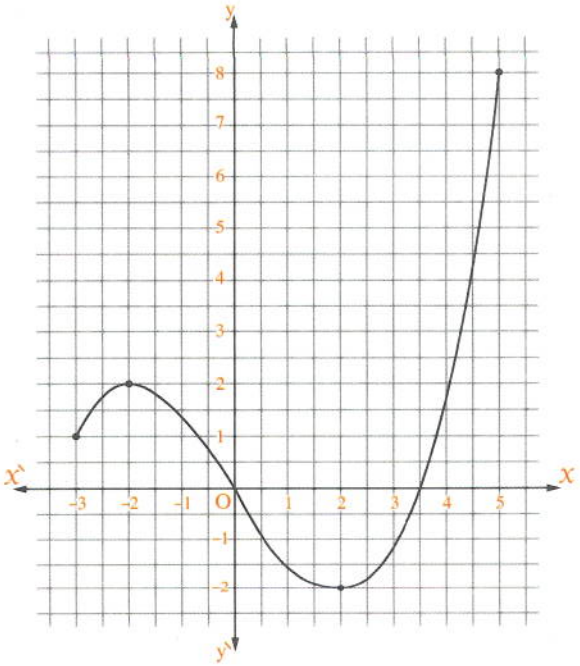
in the interval $[-5 , 4]$, then sketch the curve of the function in this interval. « $2 , -4$ »

Second Problems on the drawn curves

1 The opposite figure shows a graphical representation to the curves of the functions $f(x)$, $\grave{f}(x)$, $\acute{f}(x)$ where $f(x)$ is polynomial. Determine the curve of each function.



2 **Using the opposite figure which represents the curve of the function $f(x)$**
Choose the correct answer from those given :



(1) The function has local minimum value at $x =$ .....

(a) -3 (b) zero

(c) 2 (d) -2

(2) The absolute maximum value equals .....

(a) 5 (b) 8

(c) -2 (d) 2

(3) The curve of the function is convex upward when $x \in$ .....

(a) $]-3, 0[$ (b) $]-2, 2[$ (c) $]0, 5[$ (d) $]-3, 5[$

(4) The curve of the function has an inflection point at .....

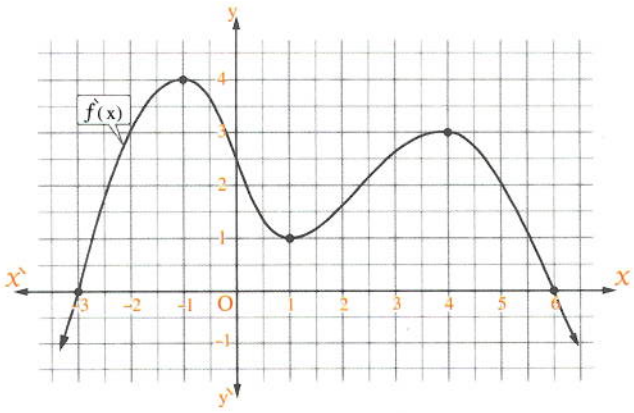
(a) $(-2, 2)$ (b) $(0, 0)$ (c) $(2, -2)$ (d) $\left(3\frac{1}{2}, 0\right)$

(5) The function is decreasing in the interval .....

(a) $]-3, 0[$ (b) $]-2, 2[$ (c) $]0, 5[$ (d) $]-3, 5[$

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3 **The opposite figure represents the curve of $\hat{f}(x)$, choose the correct answer from the given ones :**



(1) The curve $f(x)$ has local maximum value at $x =$ .....

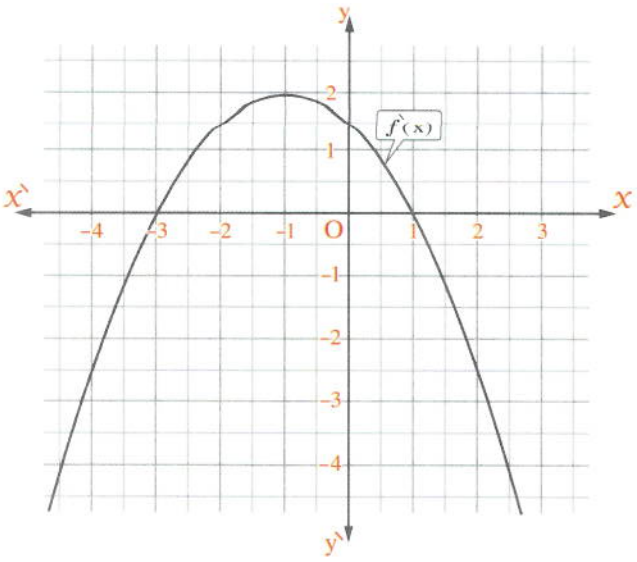
(a) -3 (b) -1

(c) 4 (d) 6

(2) The curve $f(x)$ has local minimum value at $x =$ .....

(a) -3 (b) 1 (c) 4 (d) 6

4 **The opposite figure shows the curve of $\dot{f}(x)$ for the function f where $f(x)$ polynomial :**

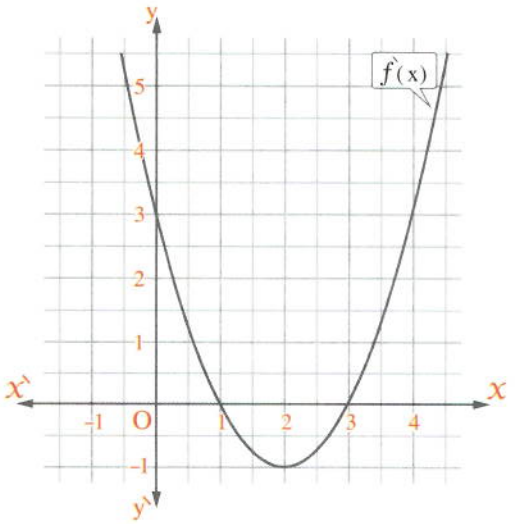


(1) Determine the increasing and decreasing intervals of the function f

(2) Find the solution set of the inequality $\ddot{f}(x) > 0$

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5 **Using the opposite figure which represents the curve of $\dot{f}(x)$. Choose the correct answer from the given ones :**



(1) $f(x)$ has local maximum value at $x =$ .....

(a) zero (b) 1

(c) 2 (d) 3

(2) $f(x)$ has local minimum value at $x =$ .....

(a) zero (b) 1

(c) 2 (d) 3

(3) $f(x)$ has an inflection point at $x =$ .....

(a) zero (b) 1 (c) 2 (d) 3

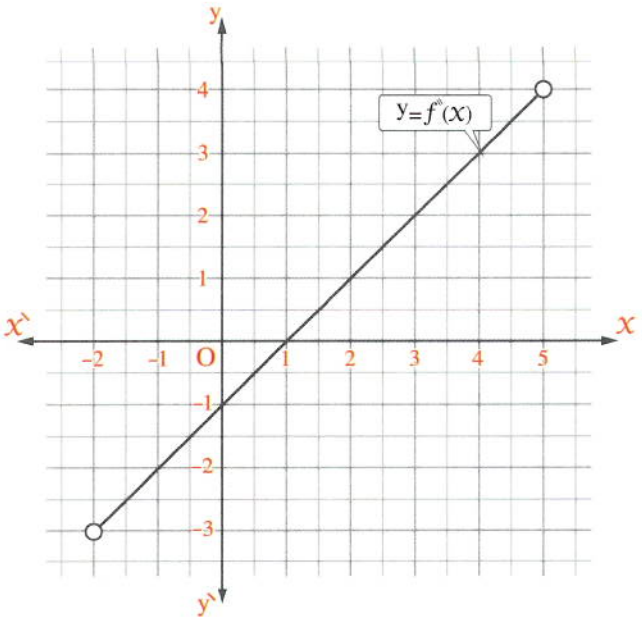
(4) $f(x)$ is convex upwards at $x \in$ .....

(a) $]-\infty , 2[$ (b) $]2 , \infty[$ (c) $]1 , 3[$ (d) $\mathbb{R}$

(5) $f(x)$ is decreasing at $x \in$ .....

(a) $]-\infty , 2[$ (b) $]2 , \infty[$ (c) $]1 , 3[$ (d) $\mathbb{R}$

6 **The opposite figure represents the curve of $f''(x)$ on the interval $]-2, 5[$ of the continuous function f :**



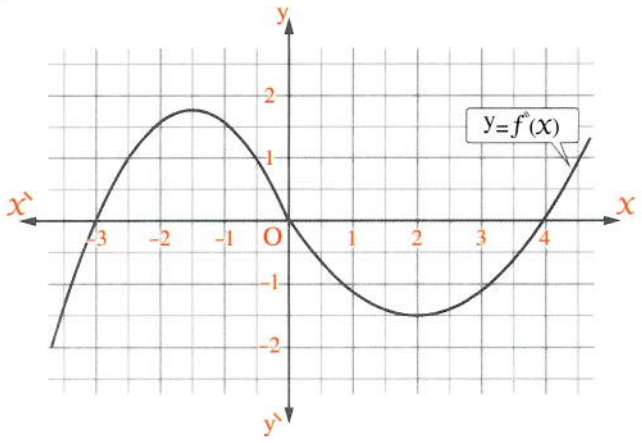
(1) Find the convexity of the intervals upwards and downwards to the curve of the function f if it is found.

(2) Are there inflection points to the curve of function f in this interval ? Explain.

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7 **Using the opposite figure that represents the curve of the $f''(x)$.**

Choose the correct answer from those given :



(1) The convex upwards interval of the function f is .....

(a) $]-\infty, 0[$

(b) $]-\infty, -3[$

(c) $]0, 4[$

(d) both (b) , (c)

(2) The convex downwards interval of the function f is .....

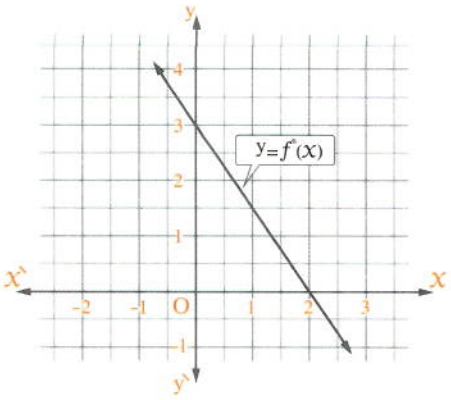
(a) $]-3, 0[$ (b) $]4, \infty[$

(c) $]0, \infty[$ (d) both (a) and (b)

(3) The point of inflection of the function f at $x =$ .....

(a) -3 (b) zero (c) 4 (d) All the previous.

8 **Using the opposite figure which represents the curve of $f'(x)$. Choose the correct answer from those given :**



(1) The curve of f is convex upwards when $x \in$ .....

(a) $]-\infty , 2[$ (b) $]2 , \infty[$

(c) $]0 , 2[$ (d) both (a) , (c)

(2) The curve of f has inflection point when $x =$ .....

(a) 1 (b) 2

(c) 3 (d) zero

(3) If $f'(-1) = f'(5) = 0$, then the function f has a local maximum value when $x =$ .....

(a) -1 (b) 2 (c) 3 (d) 5

(4) f is decreasing for each $x \in$ .....

(a) $]-\infty , -1[$ (b) $]5 , \infty[$ (c) $\mathbb{R}$ (d) both (a) , (b)

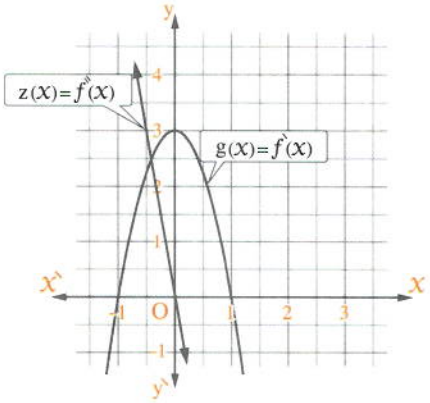
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9 The opposite figure shows the two curves of the two function g and z where :

$g(x) = f'(x)$, $z(x) = f''(x)$

and f is a polynomial function at the variable x

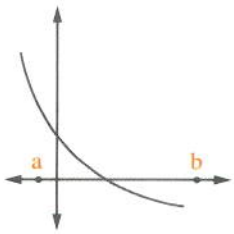
Sketch the curve of "f" known that it passes through the two points $(-1 , 0)$, $(1 , 4)$



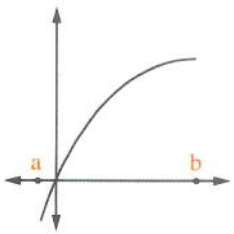
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10 **Choose the correct answer from the given ones :**

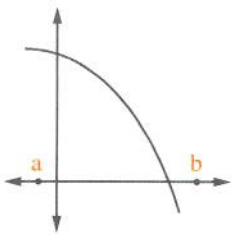
(1) If $f'(x) < 0$, $f''(x) > 0$, for each $x \in [a , b]$ state which of the following represents the curve of the function f in $[a , b]$



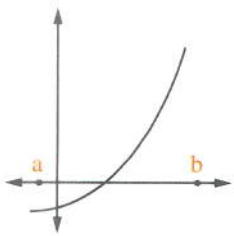
(a)



(b)

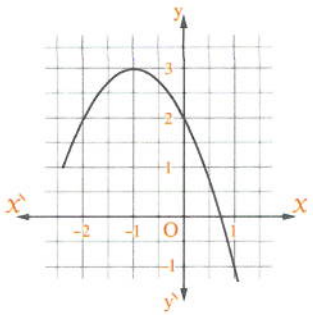


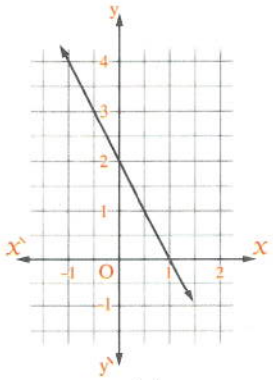
(c)



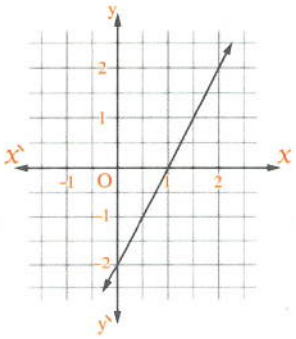
(d)

(2) The opposite figure represents the curve of the function $y = f(x)$, which of the following represents the curve $\dot{f}(x)$?

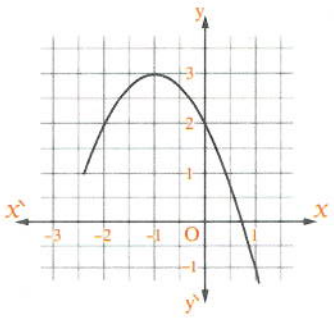




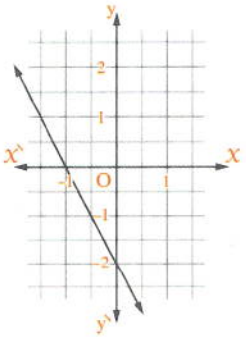
(a)



(b)

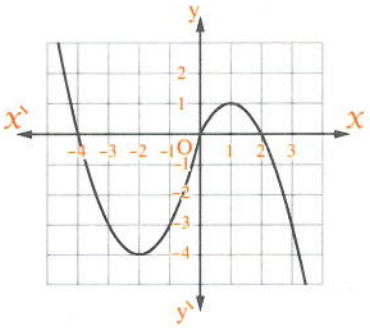


(c)



(d)

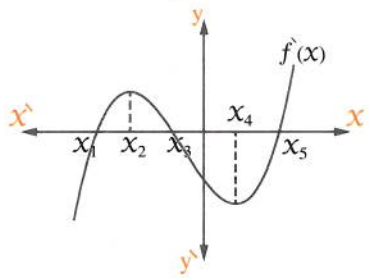
(3) The opposite figure represents the curve of the function $y = f(x)$, then all the following statements are true except .....



(a) $\dot{f}(-2) =$ zero (b) $\dot{f}(-1) > 0$

(c) $f(3) > f(5)$ (d) $\dot{f}(-4) < \dot{f}(-5)$

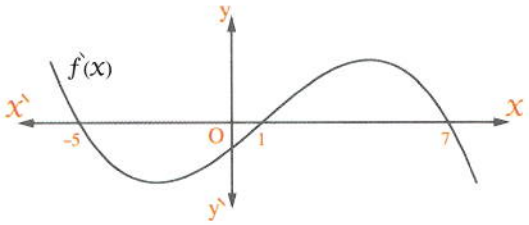
(4) The opposite figure represents the curve of the function $\dot{f}(x)$, then $f(x)$ has local maximum value at $x =$ .....



(a) x_1 , x_5 (b) x_2

(c) x_3 (d) x_4

(5) The opposite figure represents the curve $\dot{f}(x)$. All the following statements are true except .....



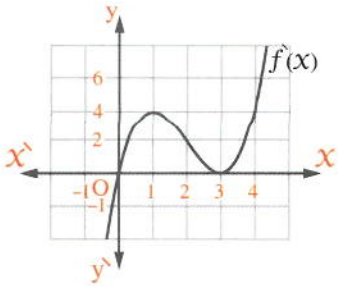
(a) at $x = -5$ the function f has a local maximum value.

(b) $\ddot{f}(1) >$ zero

(c) at $x = 7$ the function f has local minimum value.

(d) the function f is decreasing in $]-5 , 1[$

(6) The opposite figure represents the curve $f'(x)$, then the function f .....



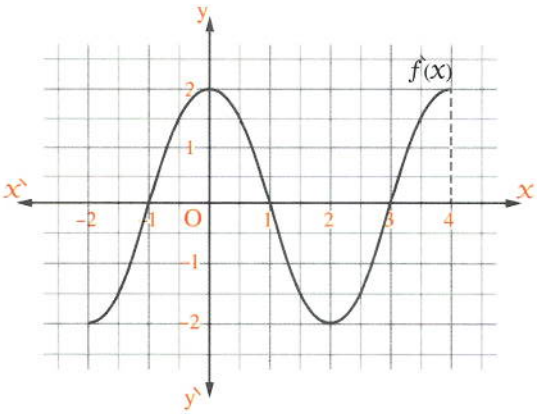
(a) has a local maximum value but not a local minimum value.

(b) has a local minimum value but not a local maximum value.

(c) has a local minimum value and a local maximum value.

(d) has neither local maximum value nor local minimum value.

(7) If the graph of $f''(x)$ (the second derivative of f) is shown in the given figure for $-2 \leq x \leq 4$, then the graph of the function f is convex upwards in .....



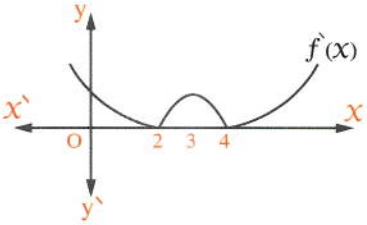
(a) $-1 < x < 1$

(b) $0 < x < 2$

(c) $-2 < x < -1$ only

(d) $-2 < x < -1$ and $1 < x < 3$

(8) The opposite figure represents the curve of the function $f'(x)$, then :



First : The function f is increasing on .....

(a) $]2 , 3[$ only (b) $]4 , \infty[$ only

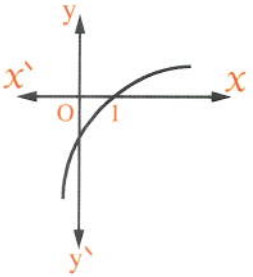
(c) $\mathbb{R}$ (d) both (a) , (b)

Second : The function f has an inflection point at .....

(a) $x = 2$ (b) $x = 3$

(c) $x = 4$ (d) All the previous.

(9) The opposite figure represents the curve of a function $y = f(x)$ which is differentiable twice , then .....



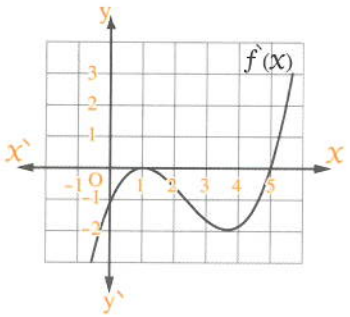
(a) $f(1) < f'(1) < f''(1)$

(b) $f''(1) < f(1) < f'(1)$

(c) $f''(1) < f'(1) < f(1)$

(d) $f(1) < f''(1) < f'(1)$

(10) The opposite figure represents the curve of the first derivative of the function $f(x)$ which of the following statements is true for sure ?



(I) $f(4) < f(3)$

(II) The function f has a local minimum value at $x = 5$

(III) The function f has a local maximum value at $x = 1$

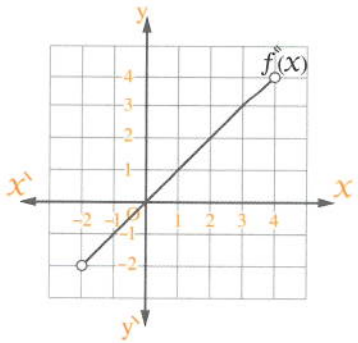
(a) only III

(b) I , II

(c) II , III

(d) only II

(11) If the opposite figure represents a continuous curve of $f''(x)$ in the interval $]-2, 4[$, then $f'(x)$ is increasing in the interval .....



(a) $]-2, 4[$

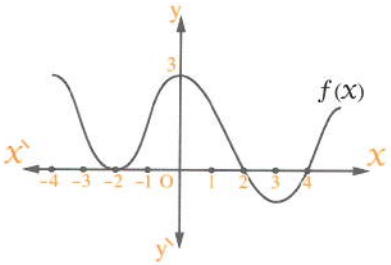
(b) $]2, 4[$ only.

(c) $]0, 4[$only.

(d) $]-2, 0[$only.

(12) The opposite figure represents the curve of the function $y = f(x)$

All the following statements are true except .....



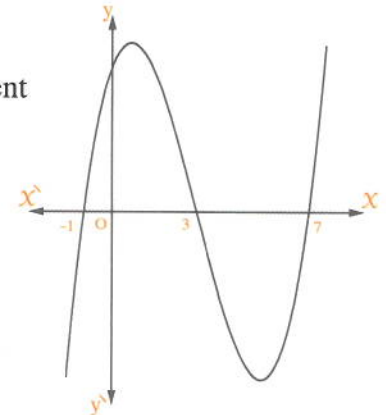
(a) $\dot{f}(-3) + \dot{f}(3) < 0$

(b) $\dot{f}(-1) > \dot{f}(1)$

(c) $\dot{f}(-2) + \dot{f}(0) = 0$

(d) $\dddot{f}(-2) < \dddot{f}(0)$

(13) If the curve of the polynomial function f has two inflection points , then the opposite figure could represent the curve of y = ...



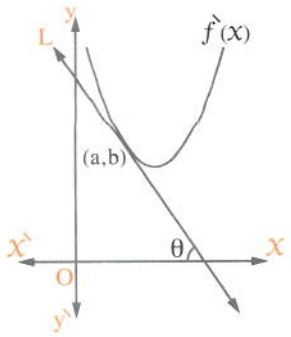
(a) $f(x)$

(b) $f'(x)$

(c) $\dddot{f}(x)$

(d) otherwise.

(14) The opposite figure represents the curve $\grave{f}(X)$, the straight line L touches the curve at (a , b) , then $\dddot{f}$ (a) = ...

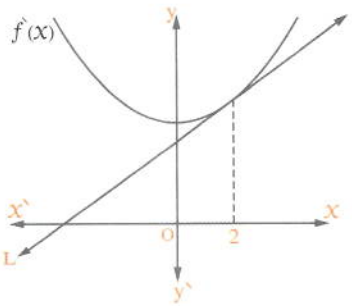


(a) tan θ (b) – tan θ

(c) zero (d) $\frac{-a}{b}$

(15) In the opposite figure :

If the straight line L : 4 y = 3 X + 12 touches the curve $\grave{f}$ (X) at X = 2 , then :



First : $\grave{f}$ (2) = ...

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$

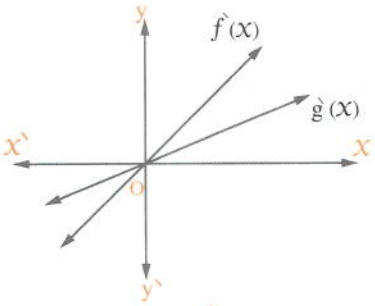
(c) 4.5 (d) 6

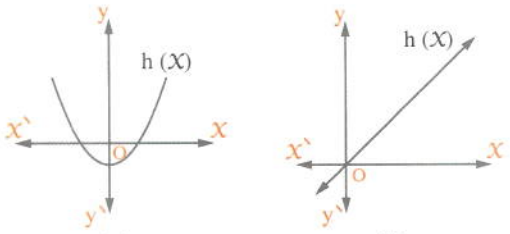
Second : $\dddot{f}$(2) = ...

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$

(c) 4.5 (d) 5

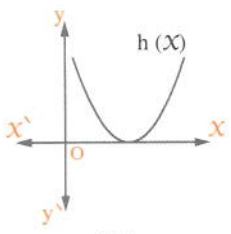
(16) The opposite figure represents the graphs of the two functions $\grave{f}$, $\grave{g}$ which of the following graphs could represent the curve of the function h where h (X) = f (X) – g (X) ?



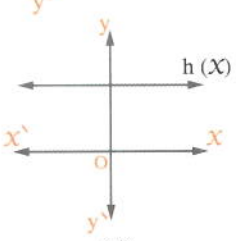


(a)

(b)

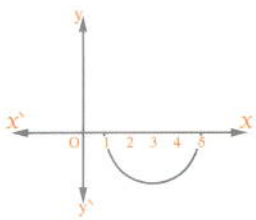


(c)

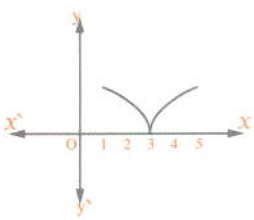


(d)

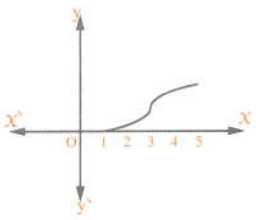
(17) ***(1st Session 2021)*** If $\grave{f}$ (3) undefined , $\dddot{f}$ (X) > 0 when X ≠ 3 , then the curve which can represent the continuous function f on [1 , 5] is .....



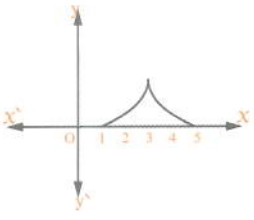
(a)



(b)

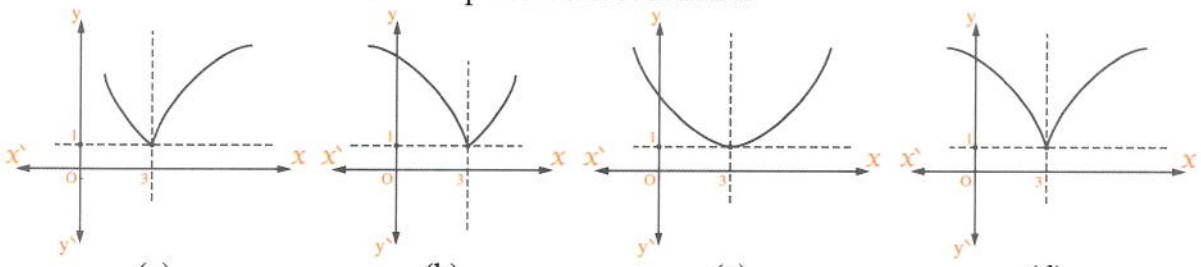


(c)



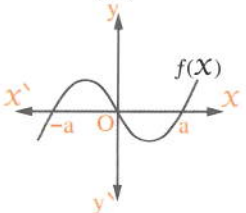
(d)

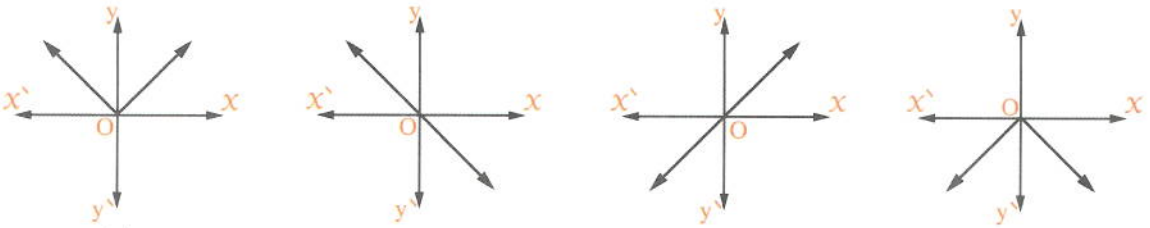
(18) *(2nd Session 2021)* If f is a continuous function has the following properties : $f(3) = 1$, $\grave{f}(x) < 0$ when $x < 3$, $\grave{f}(x) > 0$ when $x > 3$ and $\dddot{f}(x) < 0$ for every $x \neq 3$, then the curve which can represent this function is .....



(a) (b) (c) (d)

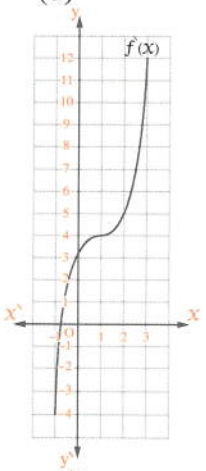
(19) *(2nd Session 2023)* If the given figure represents the curve of the function $f(x)$, then the figure could be represent the function $\dddot{f}(x)$ is .....





(a) (b) (c) (d)

(20) *(1st Session 2025)* If the opposite figure represents the curve $\grave{f}(x)$ for the polynomial function f , such that $-1 \leq x \leq 3$, then the curve of the function $f(x)$ .....



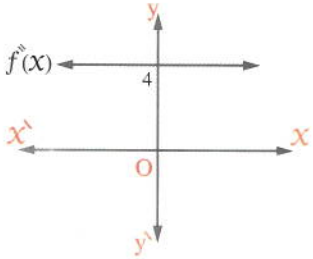
(a) has no inflection points.

(b) has one inflection points.

(c) has two inflection points.

(d) has three inflection points.

(21) *(2nd Session 2024)* If the given figure represents the curve of the function $\dddot{f}(x)$, which of the following statements is true ?



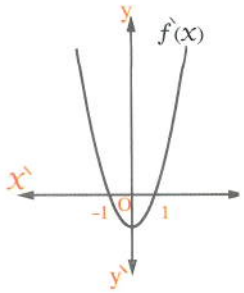
(a) The curve of the function f is convex upward in $]-\infty , 0[$

(b) The function f has an inflection point at $x = 0$

(c) The curve of the function f is convex upward in $]0 , \infty[$

(d) The function f has no local maximum value.

(22) ***(1st Session 2024)*** If the opposite figure represents the curve of $\grave{f}(x)$ of a continuous function f on $\mathbb{R}$ which of the following statements is true ?



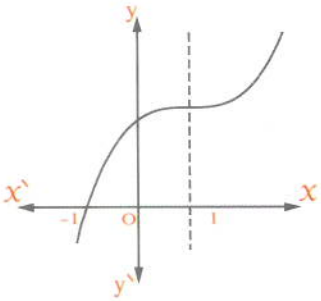
(a) The function f has a horizontal tangent at $x = 0$

(b) The function f is decreasing on the interval $]-1, 1[$

(c) The function f has a local maximum value at $x = 1$

(d) The function f has a local minimum value at $x = -1$

(23) ***(2nd Session 2021)*** The opposite figure represents the curve of the continuous function g (x) on $\mathbb{R}$ that has a horizontal tangent at $x = 1$, if f is differentiable function on $\mathbb{R}$ where $\grave{f}(x) = |g(x)|$, then the true statement from the following is .....



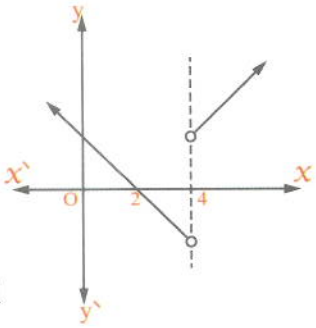
(a) The function f has inflection point at $x = -1$

(b) The function f has minimum value at $x = -1$

(c) The function f has inflection point at $x = 1$

(d) The curve of the function f convex upward in $]-1, 1[$

(24) ***(1st Session 2021)*** If the opposite figure represents the curve of the first derivative of a continuous function f whose domain is $\mathbb{R}$, then the wrong statement from the following is .....



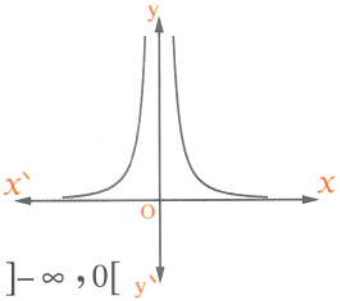
(a) the function has inflection point at $x = 4$

(b) the function has local maximum value at $x = 2$

(c) the curve of the function convex upward in $]-\infty, 4[$ and convex downward in $]4, \infty[$

(d) $f(-3) < f(-2)$

(25) ***(2nd Session 2022)*** The opposite figure represents the curve of the first derivative of the continuous function f on $\mathbb{R}$, then the wrong statement in the following is .....



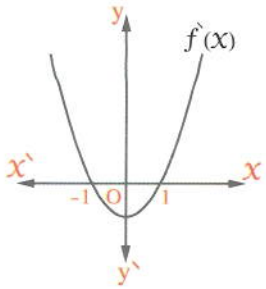
(a) The function f has an inflection point at $x = 0$

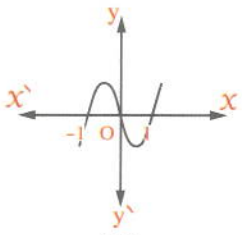
(b) The curve of the function f is convex up on the interval $]-\infty, 0[$

(c) The function f has a critical point at $x = 0$

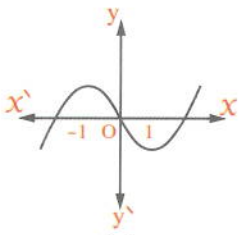
(d) The curve of the function f is convex up on the interval $]0, \infty[$

(26) The opposite figure represents the first derivative of the function $y = f(x)$ which of the following figures could be the general shape of the function $y = f(x)$?

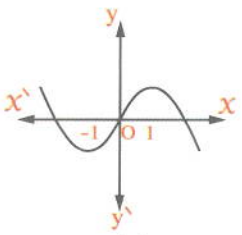




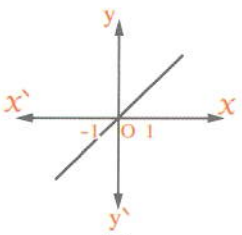
(a)



(b)

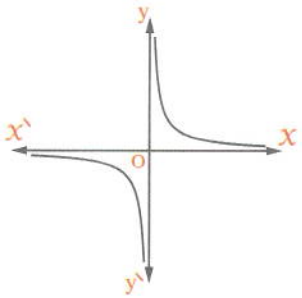


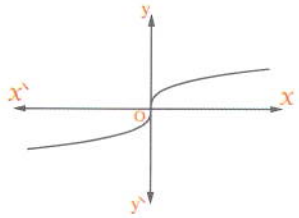
(c)



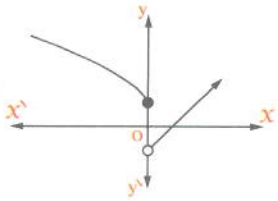
(d)

(27) ***(1st Session 2022)*** If the opposite figure represents the curve of the first derivative of the function $y = f(x)$ which is defined on $\mathbb{R}$, then the figure which may represent the curve of the function f is .....

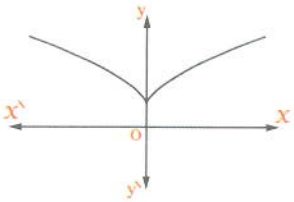




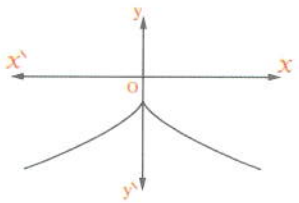
(a)



(b)

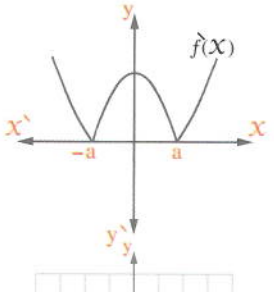


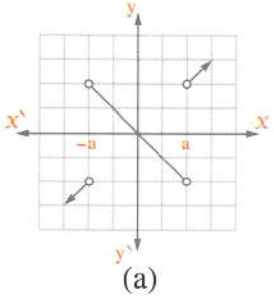
(c)



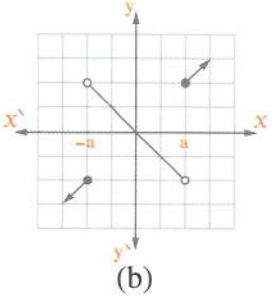
(d)

(28) ***(1st session 2023)*** The opposite figure represents the curve of the function $f'(x)$, then the figure which could represent the curve of the function $f(x)$ is .....

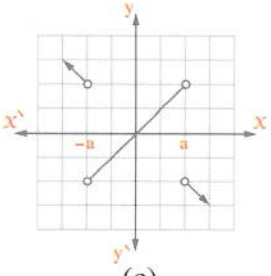




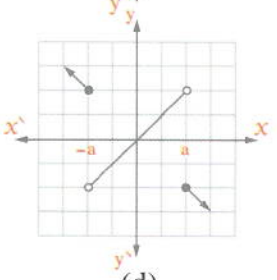
(a)



(b)



(c)

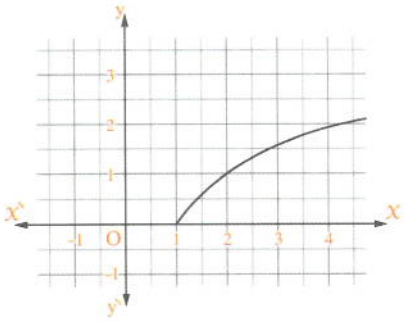


(d)

Higher skills

11 **Choose the correct answer from the given ones :**

(1) The opposite figure represents the curve of the function $y = f(x)$ If the equation of the tangent to the curve at any point (x , y) is $y = g(x)$ which of the following statements is right for each $x \geq 1$?



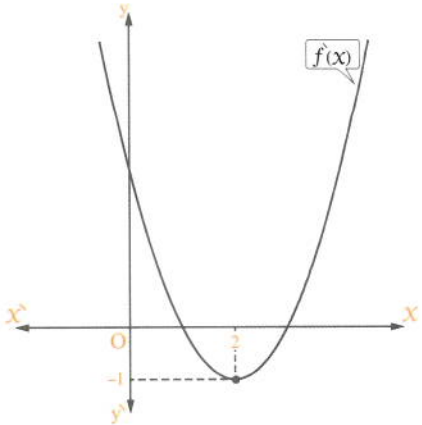
(a) $g(x) = f(x)$ (b) $g(x) \leq f(x)$

(c) $g(x) \geq f(x)$ (d) $f(x) + g(x) < 0$

(2) The opposite figure represents the curve of the derivative of the function

$f : f(x) = x^3 + 2a\,x^2 + b\,x + 1$

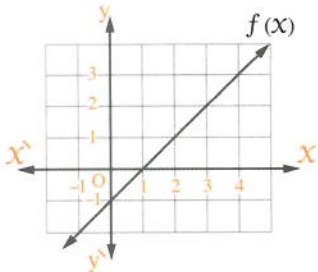
, then $f(1) =$ .....



(a) -1 (b) -3

(c) 7 (d) 11

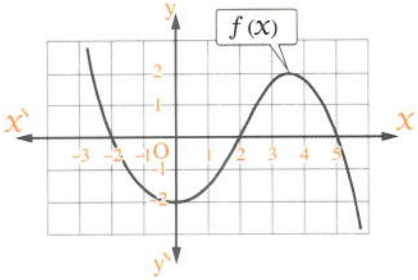
(3) The opposite figure represents the curve of the function f , then $f(x) > \dot{f}(x)$ at $x \in$ .....



(a) $\mathbb{R}$ (b) $]-\infty , 1[$

(c) $]1 , \infty[$ (d) $]2 , \infty[$

(4) The opposite figure represents the curve of polynomial function f which has an inflection point at $x = 2$, then $f(x)$, $\dot{f}(x)$, $\ddot{f}(x)$ have the same sign at $x \in$ .....



(a) $\mathbb{R}$ (b) $]-\infty , -2[$

(c) $]2 , 5[$ (d) $]5 , \infty[$